Son of Surya

GUIDE
R4
FUNDAMENTALS

FOUNDATION FOUNDATION

How a Solar Panel Works

Cells, the I-V curve, STC and real roofs, temperature effects, degradation and hot spots — read any panel datasheet with confidence.

Isc	Voc	MPP	%/°C	STC
CURRENT	VOLTAGE	POWER	HEAT	RATING

"Understand your solar before you buy your solar."

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R4 About This Guide

Guide	**R4 — How a Solar Panel Works: Cells, I-V Curve, STC, Temperature Effects**
Category · Level	Solar Fundamentals · Foundation
For	Buyers comparing panels, students, and anyone who wants to read a panel datasheet with confidence
Prerequisites	R1 Electricity Basics
Related	SOS-02 Panel Buying Guide · SOS-22 Fault Diagnosis · R3 Sun & Site

The problem

Panel datasheets are full of numbers — Pmax, Voc, Isc, Vmp, NMOT, temperature coefficients, efficiency. Buyers either ignore them or are impressed by the wrong one.

The knowledge gap

People expect a 540 Wp panel to deliver 540 W. On a hot Indian roof it usually delivers noticeably less — because of physics, not fraud.

The decision

After reading, you can explain how a panel makes electricity, read its datasheet, and judge which specifications matter for Indian heat.

The action

Take any datasheet, find the six key numbers (Section 07), and calculate the panel's output on a hot day (C2).

01 What You Will Learn

- [ ] How light becomes electricity in a silicon cell
- [ ] How cells become panels, and panels become strings
- [ ] The I-V curve: Isc, Voc and the maximum power point
- [ ] STC vs real conditions; why heat matters so much in India
- [ ] Efficiency, degradation, hot spots and bypass diodes
- [ ] The main panel technologies in plain words

02 Why This Matters

The panel is the part of your system that must work for 25 years or more. Understanding how it behaves explains why output varies through the day and the year, why shade hurts so much, why some panels do better in heat, and which datasheet promises are meaningful.

03 Fundamentals — From Sunlight to Current

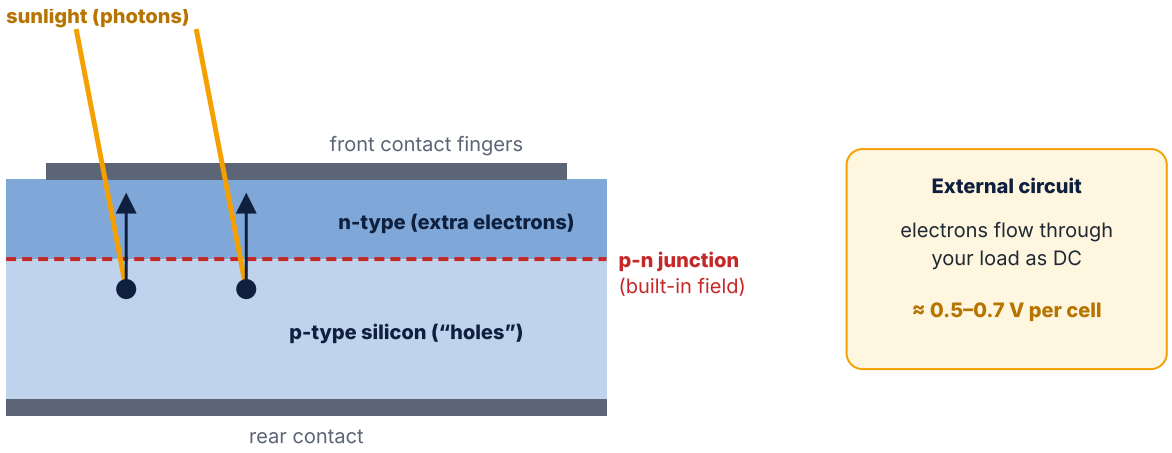


Figure 1. A silicon solar cell (simplified). Light frees electrons; the built-in field at the junction pushes them one way, creating DC current.

SIMPLE

Sunlight knocks tiny particles loose inside the silicon. A built-in slope inside the cell pushes them all in one direction — that flow is electricity.

TECHNICAL

Photons with energy above silicon’s band gap (≈1.12 eV) create electron-hole pairs. The electric field of the p-n junction separates them, producing a photocurrent proportional to irradiance and a voltage of roughly 0.5–0.7 V per cell.

PRACTICAL

Cells are connected in series inside a panel (commonly 120–144 half-cells) to reach a useful voltage; panels are connected in strings to reach the inverter’s voltage.

DECISION

Because cells in series share one current, one shaded or damaged cell can limit many others — which is why layout and quality matter.

04 Science — The I-V Curve

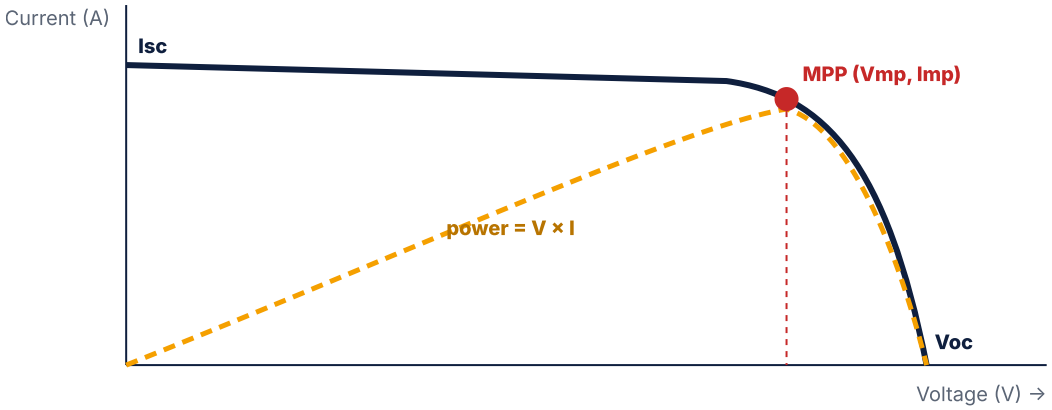


Figure 2. Current–voltage (solid) and power (dashed) curves of a panel. The inverter's MPPT keeps the panel at the maximum power point.

Point	Meaning	Why it matters
Isc	Short-circuit current — maximum current, at zero voltage	Sets cable, fuse and inverter input current limits
Voc	Open-circuit voltage — maximum voltage, at zero current	Sets the maximum string length (rises in cold)
MPP (Vmp, Imp)	The point where V × I is highest = Pmax	Where the inverter tries to operate

How conditions move the curve

- **Less sunlight** → current falls roughly in proportion; voltage changes little.
- **Higher temperature** → voltage falls noticeably; power falls by the **temperature coefficient of Pmax** (typically about −0.25% to −0.40% per °C, depending on technology — check the datasheet).
- **Shade on some cells** → the curve gets steps; bypass diodes switch in to protect the shaded section.

05 How It Works in Practice — STC, NMOT and Real Roofs

Rating condition	Definition	Use
STC	1,000 W/m² irradiance, 25 °C cell temperature, AM 1.5 spectrum	The "Wp" on the label; for comparing panels
NMOT / NOCT	Nominal operating conditions: lower irradiance (800 W/m²), real ambient temperature and wind; gives a realistic cell temperature	Closer to real-world output

On a sunny Indian roof, cells commonly run **far above 25 °C**. The datasheet NMOT value hints how hot your panel will run in normal operation; higher ambient air, low wind and poor airflow push it higher.

WHAT CHANGES IN INDIA?

High ambient temperatures for much of the year mean the **temperature coefficient** matters more in India than in cooler countries. Two panels with the same Wp can produce different yearly units on the same roof because one handles heat better. Leave airflow under panels; avoid mounting them flat on hot surfaces.

06 Indian Context — Ageing, Heat, Dust and Damp

Effect	What happens	What to check
Degradation	Output declines slowly over years	Performance warranty curve (SOS-02)
Hot spots	A shaded, cracked or dirty cell heats up and can damage the panel	Avoid partial shade; clean droppings; thermal check if output drops
Bypass diodes	Let current skip a shaded cell group	Usually 3 per panel, in the junction box
PID	Potential-induced degradation — voltage stress plus humidity can reduce output	PID-resistance testing on the datasheet; relevant in humid coastal areas
Soiling	Dust, pollen, bird droppings block light	Cleaning plan (SOS-20)
Salt mist, ammonia	Corrosive environments near the sea or livestock sheds	Salt-mist (IEC 61701) and ammonia (IEC 62716) test certificates

Panel technologies in plain words

Term	What it means
Mono-crystalline	Cells cut from a single silicon crystal; the mainstream today
PERC	A rear-surface cell design that improved efficiency; widely used
TOPCon, HJT	Newer cell structures, often with better temperature coefficients — compare datasheets
Half-cut cells	Cells cut in two to reduce current per cell and resistive losses; can tolerate partial shade better
Bifacial	Collects light on the back too; gains depend on mounting height and a reflective surface below

Technologies change quickly. Judge a panel by its **datasheet, certifications and warranties**, not by the technology name alone.

07 Practical Application — The Six Datasheet Numbers

#	Number	What to look for
1	Pmax (Wp) and tolerance	Rated power; positive-only tolerance (e.g. 0 to +5 W) is better
2	Module efficiency (%)	Higher = more watts per m² (matters on small roofs)
3	Temperature coefficient of Pmax (%/°C)	Closer to zero is better in Indian heat
4	Voc, Isc, Vmp, Imp	For the designer's string and inverter checks
5	NMOT (°C)	Lower suggests cooler operation
6	Dimensions and weight	Layout and roof loading

08 Calculations

C1 Module efficiency

FORMULA	$\eta = P_{max} \div (\text{Area} \times 1{,}000\ W/m^2)$
ASSUMPTIONS	540 Wp; 2.278 m × 1.134 m (example dimensions)
WORKING	540 ÷ (2.583 × 1,000) = **20.9%**

C2 Output on a hot day

FORMULA	$P = P_{max} \times [1 + \gamma \times (T_{cell} - 25)] \times (G \div 1{,}000)$
ASSUMPTIONS	540 Wp; cell temperature 60 °C; G = 1,000 W/m²; γ = −0.35 %/°C (panel A) or −0.29 %/°C (panel B) — example values
WORKING	A: 540 × (1 − 0.0035 × 35) = **473.9 W** · B: 540 × (1 − 0.0029 × 35) = **485.2 W**
INTERPRETATION	Same label, ≈ 11 W (2.4%) difference at that moment. Over many hot hours, a better coefficient adds up — part of why datasheets matter more than labels.

C3 Less sunlight

WORKING	At 600 W/m² (25 °C cells): 540 × 0.6 = **324 W**
INTERPRETATION	Output follows sunlight closely — morning, evening and cloudy hours produce much less.

C4 Voltage of a string	
WORKING	10 panels, Voc 49.5 V each → 495 V at STC; at 5 °C with β −0.27 %/°C: 49.5 × 1.054 × 10 = **521.7 V**
INTERPRETATION	Strings reach hundreds of volts DC — professional-only territory (R1).

09 Indian Case Study

ILLUSTRATIVE EXAMPLE — not a real customer or product

A homeowner in Nagpur compared two 540 Wp panels at similar prices. Panel A: γ −0.35 %/°C, 0 to +3 W tolerance. Panel B: γ −0.29 %/°C, 0 to +5 W tolerance, PID-resistance certificate. Nagpur summers are among the hottest in India. Using C2, the owner saw B produce ≈ 2–3% more in the hottest hours. With equal warranties and similar prices, B was the better fit for the climate.

10 Cost & Economics

Panel price is usually compared per Wp. But two panels with the same price per Wp can deliver different **kWh per Wp** in hot climates, and different output after 20 years. A small price premium for a better temperature coefficient or slower degradation can pay back over the system's life — the panel buying guide (SOS-02) shows how to compare.

11 Common Mistakes

Mistake	Reality
Expecting the full Wp all day	Wp is a lab rating at 25 °C
Ignoring temperature coefficient	It matters in Indian heat
Thinking efficiency = more units per Wp	Efficiency means less area per Wp, not more kWh per Wp
Treating a little shade as harmless	Series cells and hot spots make it costly
Choosing by technology name	Compare datasheets and warranties

12 Buyer Protection

- [ ] Ask for the datasheet of the exact model quoted
- [ ] Check the label on delivered panels matches the datasheet and quote
- [ ] Ask for BIS registration and ALMM listing of the model (SOS-02)

- [ ] In coastal or livestock areas, ask for salt-mist or ammonia test certificates
- [ ] Ask for PID-resistance information in humid regions

13 Practical Checklist — Reading a Datasheet

- [ ] Pmax _____ Wp · tolerance _____
- [ ] Efficiency _____ %
- [ ] γ (Pmax) _____ %/°C · β (Voc) _____ %/°C
- [ ] Voc _____ V · Isc _____ A · Vmp _____ V · Imp _____ A
- [ ] NMOT _____ °C · dimensions _____ · weight _____
- [ ] Certifications listed · warranty terms attached

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Save your panels' datasheets and serial numbers in the free **Solar Project Passport** — you will need them for every warranty claim.

14 Problem Solving

Level	Problem	Solution
EASY	A 400 Wp panel at 500 W/m² and 25 °C?	400 × 0.5 = **200 W**
INTERMEDIATE	550 Wp, γ −0.30 %/°C, cells at 55 °C, 1,000 W/m²?	550 × (1 − 0.003 × 30) = **500.5 W**
REAL-WORLD	A 580 Wp panel is 2.278 × 1.134 m. Efficiency? Is it more units per Wp than a 540 Wp panel of the same size?	580 ÷ 2,583 = **22.5%**. Not necessarily — it gives more watts per panel; kWh per Wp depends on heat behaviour, losses and site.

15 FAQ

Q1 Do panels work in winter?
Yes — cooler cells are more efficient, though days are shorter.

Q2 Do panels work under moonlight or streetlights?
Negligibly.

Q3 Are black panels better?
Colour is mostly appearance; compare datasheets.

Q4 Does hail break panels?
Panels are tested for hail impact; severe events can still damage them.

Q5 What is a bypass diode?
A small device that lets current skip a shaded cell group.

Q6 Is bifacial worth it on my roof?
Usually little gain on flat roofs with low mounting height.

Q7 How long do panels last?
Designed for decades; see the performance warranty.

Q8 Do cracks matter?
Yes — micro-cracks can create hot spots. Handle panels carefully.

16 Glossary

Band gap — energy needed to free an electron (Si ≈ 1.12 eV).

Bifacial — collects light on both sides.

Bypass diode — protects shaded cell groups.

Hot spot — overheated cell area.

Isc — short-circuit current.

MPP / MPPT — maximum power point / tracking.

NMOT — nominal module operating temperature.

PID — potential-induced degradation.

p-n junction — boundary that creates the cell's field.

STC — standard test conditions.

Temperature coefficient — change per °C.

Voc — open-circuit voltage.

17 What To Do Next

R4 ✓ → SOS-02 Panel Buying → SOS-03 Inverters → Passport

18 Sources & Disclaimer

Sources: IEC 60904-3 (STC); IEC 61701 (salt mist) and IEC 62716 (ammonia) test standards; standard photovoltaic physics (band gap, I-V characteristics); MNRE quality control order (IS 14286, IS/IEC 61730) via pv magazine India. Checked 7 Oct 2026.

Disclaimer: Educational content by Son of Surya. Example values (dimensions, coefficients) are illustrative — use the actual datasheet. Examples marked ILLUSTRATIVE EXAMPLE are invented. No product or brand is endorsed. Electrical work is for qualified professionals only.

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India’s structured solar knowledge library

Free foundation guides, practical calculators and honest, step-by-step knowledge — so every Indian home, farm and business can understand solar before buying it.

UNDERSTAND → CALCULATE → VERIFY → COMPARE → DECIDE → ACT

FREE TOOL

Son of Surya Solar Calculator

Enter your bills and location. See system size, yearly units and savings — with every assumption shown.

FREE TOOL

Solar Project Passport

Keep your solar project’s key numbers, documents and warranties in one place. An engineer-verified version is available.

FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

SPECIALISED GUIDES — RELEASING IN BATCHES

- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
- **Professional** — wiring, installation, EPC

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Web sonofsurya.com **Instagram** @son.of.surya._ **WhatsApp** +49 160 4923611

Email sonofsurya15@gmail.com